GrandCode: Achieving Grandmaster Level in Competitive Programming via Agentic Reinforcement Learning

The paper introduces *GrandCode*, a multi-agent reinforcement-learning system that surpasses top human competitors in live Codeforces contests. GrandCode orchestrates several specialized agents— a main solver, a hypothesis generator, a summarization model, and a test-case generator—through a tightly coupled RL loop. Central to its training is *Agentic GRPO*, an extension of Group Relative Policy Optimization that provides immediate reward updates and delayed corrections, addressing off-policy drift in long, multi-stage rollouts. The system undergoes extensive pipeline stages: continued pre-training on noisy competitive-programming data, supervised fine-tuning with verified reasoning traces, and multi-component RL that jointly optimizes the solver, hypothesis, and summarizer. Test-case generation combines difference-driven and solution-attack strategies, while online test-time RL (with a rank-based reward smoothing) fine-tunes the main solver during live contests. Empirical results show GrandCode achieving first place in three recent Codeforces Rounds (1087-1089), outperforming all human participants, including grandmasters, and achieving high accept rates and weighted scores across benchmarks. The paper also details system architecture, asynchronous training with pipelined context parallelism, dynamic difficulty-aware batching, and comprehensive evaluations of each component.

Verification.
https://arxiv.org/html/2604.02721v2